Saturday
12 September 2026

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Quantum Physics course – coursera

- Moored Law : Empirical observation & projection starting that the number of transistors on a microscopic doubles roughly every 2 years, while cost of computers is halved.

✡ IBM Q cryostat used to keep IBM's 50 qubit computer cold in the IBM Q lab in Yorktown heights

✡ Quantum Revolution : Quantum Computing
Quantum Sensing & Imaging
Quantum Communications.
Quantum Information Science &

→ Specialization Outline

1) Course-1 : Foundations of Quantum Mechanics
2) Course-2 : Theory of Angular Momentum
3) Course-3 : Approximation Methods

→ Course-1 Foundations of Quantum Mechanics

- Wave particle duality & Schrodinger Eq^n
- 1-D potential problems : Potential well, potential barrier & harmonic oscillator.
- Operators & Measurements, how QM predicts the results of measurements.
- Time evolution of Quantum states.

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- Ensemble of identical particles

→ Course-2 : Theory of Angular Momentum

- Orbital angular momentum & hydrogen atom.
- Rotation & Angular Momentum.
- General theory of angular momentum

→ Course-3 : Approximation Methods

- Time-independent perturbation theory
- Time-dependent perturbation theory
- Other approximation methods

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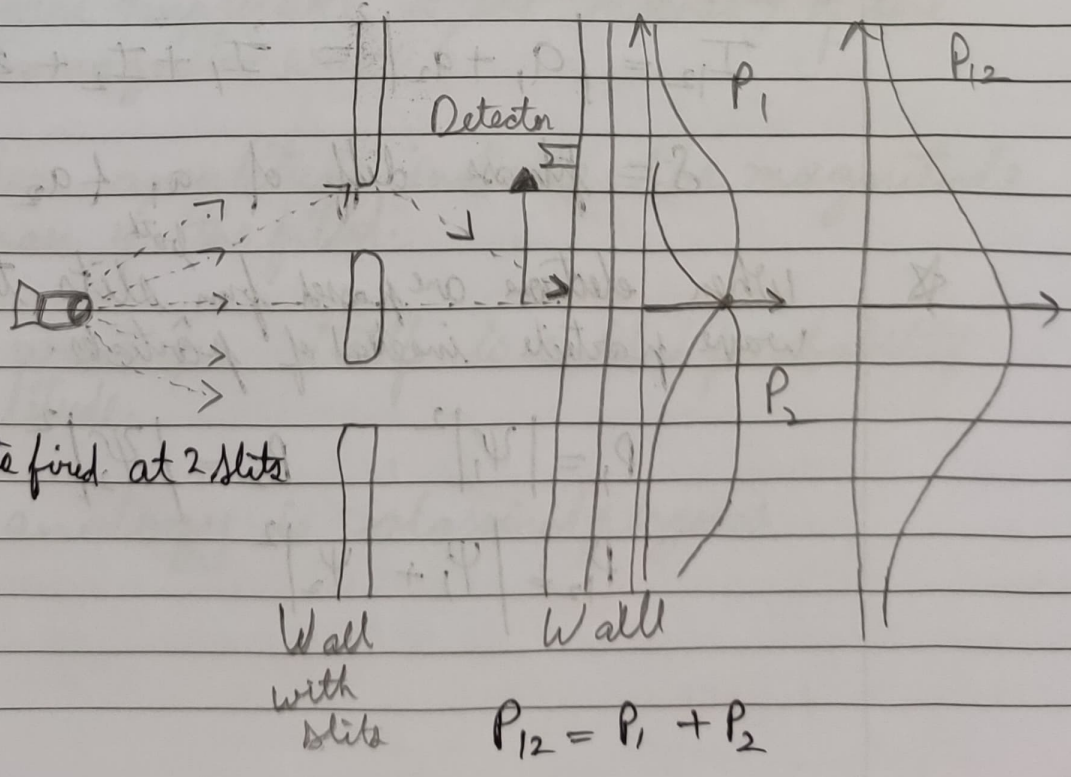
* Course - 1 Foundations of Quantum Mechanics
Course Introduction

→ Course outline

1) Wave particle duality and Schrödinger Eq^n
2) 1-D potential problems
3) Operators & measurements - 1 & 2
4) Time evolution of quantum states
5) Ensembles & identical particles.

* 1) Wave particle duality — 10

- Double slit Experiment



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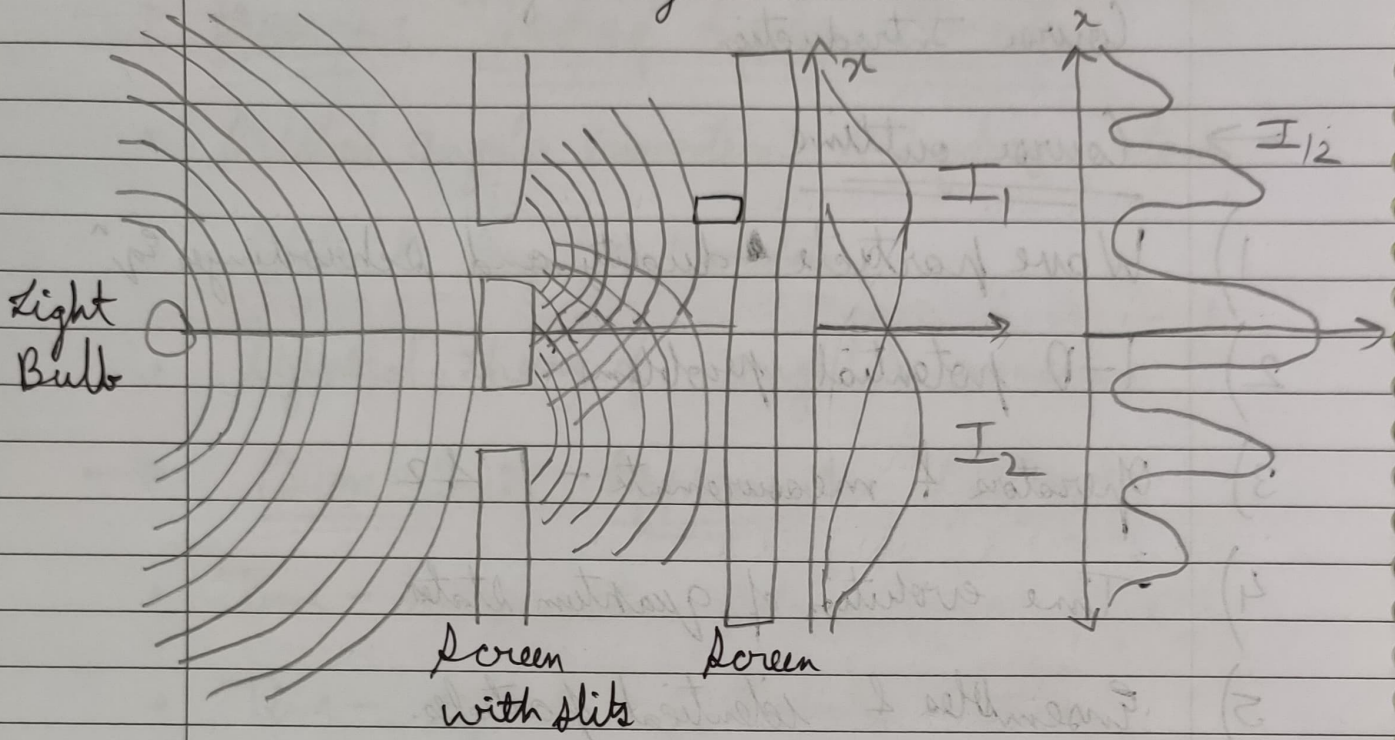
→ Wave source — Light Bulb



$$I_1 = |a_1|^2 \qquad I_2 = |a_2|^2$$

$$I_{12} \neq I_1 + I_2$$

$$I_{12} = |a_1 + a_2|^2 = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta$$

δ = phase diff of a_1 & a_2

☆ When electrons are passed from both slits they behave as wave particle instead of particles.

$$P_1 = |\psi_1|^2 \qquad P_2 = |\psi_2|^2$$

$$P_{12} = |\psi_1 + \psi_2|^2$$

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→ In Quantum Physics, the cat experiment of Schrodinger, the cat is alive & dead at the same time until observed. The cat has 2 probability wave fns active at that moment of time, but as the cat is observed the wave fn collapses & chooses only one specific reality & the one probability wave fn.

→ The concept of a quantum ~~computing~~ system existing in a hybrid state of multiple possibilities simultaneously (superposition) until it is observed is the fundamental building block of a quantum computer.

→ ~~Instead of~~ A quantum computer uses the unobserved hybrid state to process multiple possibilities at the exact same time.

1) Superposition : the Ultimate Multitasker

In a classical computer, a bit is a physical switch that is either a 1 (alive) or 0 (dead). If you have 3 classical bits, they can only hold upto one specific combination at a time like (1-0-1). To test different combinations, the computer has to run through them one by one.

✡ A qubit (quantum bit), acting like Schrodinger cat, isn't forced to pick 0 or 1 until observed. If we put 3 qubits in superposition they exist in all 8 possible combinations (2^n) $(2^3 = 8)$

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simultaneously. The computational space doubles every qubit you add.

2) Quantum gates & Interference

- Quantum gates act ~~both~~ like wave manipulators.
- Think of throwing 2 stones into a pond. The ripples will cross each other. When 2 peaks meet, the wave gets bigger (constructive interference). When a peak meets a trough, they cancel out (destructive interference)
- Because unobserved qubits act as probability wave fns, passing them through quantum gates manipulates those waves. A quantum algorithm is designed so that the wrong answers destructively interfere — cancel out each other & correct answer constructively interfere (amplifies itself).

3) Observation — Wave fn collapse

- Just like opening the box to check on the cat, we eventually have to read an answer. When we "measure" the qubits at the end of the circuit, the superposition collapses. Because the quantum gates amplified the probability of the correct path, the wave fn almost certainly collapses exactly onto the solution we need.

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✱ Key: A quantum computer saves time not by pushing more energy or particles through a circuit but by entering a hybrid state where it can look at every possible answer at once, cancel out the wrong ones, & collapse onto the right one.

I) Classical vs Quantum Information

The foundational differences lie in how information is stored & processed.

- Classical bit: A deterministic physical state.
  - State: $x \in \{0, 1\}$
  - It is mutually exclusive. It is either 0 or 1
- Quantum Bit (Qubit): A probabilistic state vector residing in a 2-D complex vector space (Hilbert space)
  - Basis States: Denoted by Dirac Notation (Bra-Ket)
    - $|0\rangle$ Equivalent to classical 0, spin up or ground state
    - $|1\rangle$ Equivalent to classical 1, spin down or excited state
  - Vector representation

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

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✱ V Vector Space

1) for all $\underline{\Psi}, \underline{\Phi} \in V$, then

$$\underline{\Psi} + \underline{\Phi} \in V$$

2) for scalar c, then $c \cdot \underline{\Psi} \in V$

3) Also, $\vec{0} \in V$

Technical definition for a vector space is that it is a set V, over a field F, equipped with a 0 element & 2 operations (addition & scalar multiplication) satisfying the following axioms:

- VS1: $\forall u, v \in V, u + v = v + u$
- VS2: $\forall u, v, w \in V, (u+v) + w = u + (v+w)$
- VS3: $\forall u \in V, u + 0 = u$
- VS4: $\forall u \in V$, there is some $v \in V$: $u + v = 0$
- VS5: $\forall u \in V, 1 \cdot u = u$
- VS6: $\forall u \in V, a, b \in F, (ab)u = a(bu)$
- VS7: $\forall u, v \in V, a \in F, a(u+v) = au + av$

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- VS8: $\forall u \in V, a, b \in F, (a+b)u = au + bu$
- $[u_1, u_2, u_3, \ldots\ldots, u_n]$ live in a n-dimensional vector space row vector space.
- $\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ live in a n-dimensional column vector space
- $\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{12} & \cdots & a_{nn} \end{bmatrix}$ live in a n^2-dimensional vector space all elements of $n \times n$ matrices
- For $X \subset \mathbb{R}$, then space of all fnas $f: X \to \mathbb{R}$ form a vector space

→ Hilbert space $\vec{u}, \vec{v}$ → abstract vectors
dot product → inner product.

→ Hilbert space is vector space that is
1) Complete
2) Inner product.

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4) Complete

Polynomial fn is any fn written in the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x^1 + a_0$$

where $a_n, a_{n-1}, \ldots, a_1, a_0$ are real nos

Polynomials form a vector space, $P(\mathbb{R})$ with basis.

$$\{x^0, x^1, x^2, x^3, x^4, x^5, \ldots\}$$

Any function in Hilbert space can be written as a linear combination of these powers.

Example
$P(x) = \sin x$
$P(x) = e^x$
$P(x) = \cos x$

$\lim_{n \to \infty}$ in all 3 cases by using taylor series expansion.

* When limit goes to ∞ the function ~~too~~ changes and a new fn appears

Topological Vector spaces

Groups

Rationals
3.1415...

Open Intervals
(π) Irrational

Fn spaces

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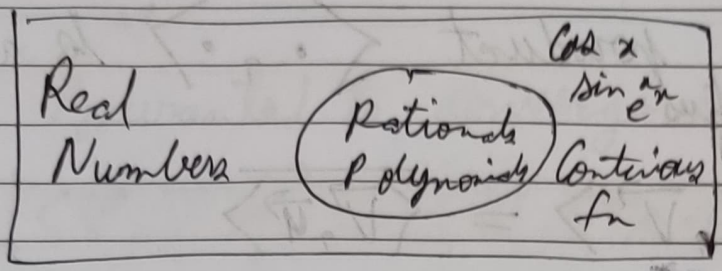
Part of larger space which completes smaller one



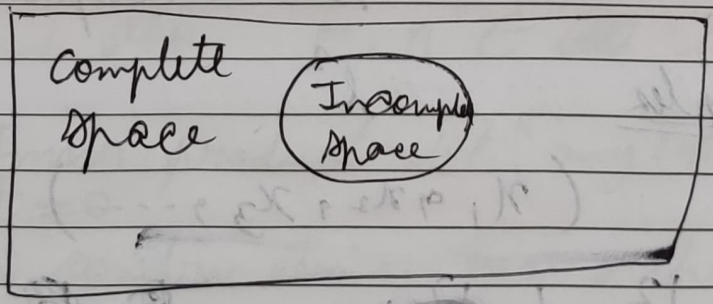
→ These type of sequences are called cauchy sequences.

✱ A sequence $x_1, x_2, \ldots$ such that for every

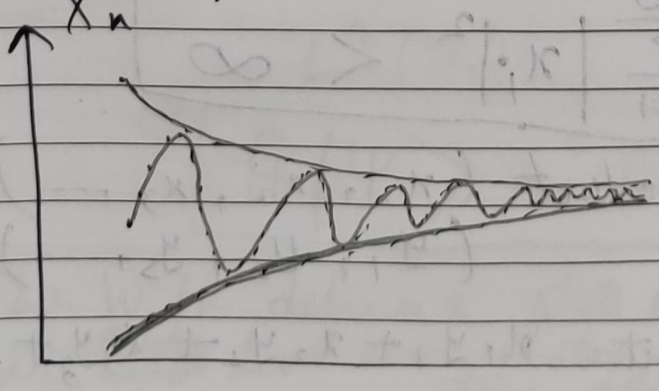
$$\varepsilon > 0 \quad \exists M \in \mathbb{N}$$

such that $m, n \geq M$ then

$$d(x_m, x_n) < \varepsilon \Rightarrow |x_m - x_n| < \varepsilon$$



→ Vector Space + Inner Product



No matter the space

some notion of distance is needed

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✡ Inner Product

An inner product $\langle \cdot, \cdot \rangle$ is a fn that satisfies:

$$\langle \vec{u}, \vec{v} \rangle = \overline{\langle \vec{v}, \vec{u} \rangle}$$

$$\langle a\vec{u} + b\vec{v}, \vec{w} \rangle = a\langle \vec{u}, \vec{w} \rangle + b\langle \vec{v}, \vec{w} \rangle$$

$\langle \vec{v}, \vec{v} \rangle \geq 0$ with $\langle \vec{v}, \vec{v} \rangle = 0 \iff \vec{v} = 0$

→ The dotproduct $\vec{u} \cdot \vec{v}$ is an inner product $\langle \vec{u}, \vec{v} \rangle$ on $\mathbb{R}^n$

→ Hilbert space = Inner product space that is complete.

- Examples l^2

$$(x_1, x_2, x_3, \dots)$$

$$|x_1|^2 + |x_2|^2 + \dots < \infty$$

$$\boxed{\sum_{i=1}^{\infty} |x_i|^2 < \infty}$$

Inner product $(x_1, x_2, x_3 \dots)$
$(y_1, y_2, y_3, \dots)$

Inner product $= x_1 y_1 + x_2 y_2 + x_3 y_3 + \dots$

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$$\text{Inner Product} = \sum_{i=1}^{\infty} x_i y_i$$

Guaranteed to converge by Cauchy Schwartz

$$|\langle \vec{u}, \vec{v} \rangle|^2 \leq \langle \vec{u}, \vec{u} \rangle \cdot \langle \vec{v}, \vec{v} \rangle$$

Example 2 $L^2[0, 2\pi]$

$$\int_0^{2\pi} |f(x)|^2 \, dx < \infty$$

$$\langle f, g \rangle = \int_0^{2\pi} f(x) \cdot g(x) \cdot dx$$

- Orthogonal basis perpendicular to each other $\{\sin(nx), \cos(nx)\}$ $n, n \in$ integers
- Inner product $\int$ any 2 fns in above set $= 0$
- certain combinations form orthonormal basis

Example-3 Schrodinger Wave fn in QM

$L^2(\mathbb{R})$

$$\hat{H}|\psi\rangle = E|\psi\rangle$$

The W fn defines a state of a physical system & it lives in a related Hilbert space this time defined over all real nos.

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$$\int_{\mathbb{R}} |\psi(x)|^2 \, dx < \infty$$

Rilmann Integral

→ There will be sequences that converge to something outside of the space. In order for this space to be complete and thus actually be a Hilbert space the Integral there needs to be more versatile Lebesque Integral

$$\int_{\mathbb{R}} |\psi(x)|^2 \, d\mu < \infty$$

Lebesque Integral

II) Superposition — The Schrodingers Cat State

- Superposition is the ability of a quantum system to be in multiple states at the same time until observed. This is the mathematical equivalent of the cat being "alive & dead".
- The Wave fn (ψ) : A qubit in superposition is described by a linear combination of its basis states

Equation $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

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- The amplitudes (α & β)
  - These are complex nos. representing probability amplitudes.
  - They are not the probabilites themselves.
- ~~Probabilities~~ (α & β) Born Rule (Probability Calculation)
  - Probability of measuring the state $|0\rangle$ is the magnitude squared of its amplitude:

$$\boxed{P(0) = |\alpha|^2}$$

  - Probability of measuring the state $|1\rangle$

$$\boxed{P(1) = |\beta|^2}$$

- Normalisation ~~constant~~ Constraint:

Because the qubit must collapse to either 0 or 1 upon measurement the total probability must be equal to 100% or 1

$$\boxed{|\alpha|^2 + |\beta|^2 = 1}$$

$$P(0) + P(1) = 1$$

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III) Quantum Gates (Waves manipulation)

- Quantum Computers do not use "gates" in the classical sense of blocking or allowing physical current. They use mathematical operations to manipulate the probability amplitudes (α & β) of the wave fn without measing it.

- Quantum gates are represented by Unitary matrices U when a gate acts on a qubit, it is multiplying a matrix by a vector.

$$\boxed{\text{Equation: } |\psi_{new}\rangle = U|\psi_{old}\rangle}$$

- Key example. Hadamard gate (H)

- This is the gate that creates the "Schrodinger's Cat" state. It takes a deterministic basis state and puts it into a perfect 50/50 superposition.

$$\text{Matrix } H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Operation on $|0\rangle$:

$$H|0\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

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→ Result: the probabilities are $\left|\frac{1}{\sqrt{2}}\right|^2 =$

$\frac{1}{2}$ for $|0\rangle$ & $\frac{1}{2}$ for $|1\rangle$. A perfect coin flip

IV) Computation via Interference

This is how quantum computers achieve speedups. It is not about raw speed or parallel physical circuits, it is about wave mechanics.

1) Initialization: Qubits begin in a known state usually $|0\rangle$.

2) Superposition: Hadamard gates put qubits into superposition, allowing the system to represent all possible inputs simultaneously.
n qubits = 2^n simultaneous states.

3) Algorithmic manipulation — Interference:

- The quantum algorithm applies a sequence of gates (matrices) to the superposed state.
- Constructive Interference: Amplitudes leading to correct answers are mathematically amplified they combined grow larger.

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- Destructive Interference: Amplitudes leading to incorrect answers are mathematically cancelled out, a positive & a −ve amplitude sum to zero.

4) Measurement — Wave fn collapse

- Observation forces the wave function to collapse form a superposition of many states into a single classical state.

- Because of the interfered steps, the probability ($|\alpha|^2$ or $|\beta|^2$) of collapsing into the correct answer is close to 100%.

Case-1 Gate matrix (H) State $|1\rangle$ Result Vector

Hadamard

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{bmatrix}$$

$|\psi|^2$ → Prob $|0\rangle = 50\% = \frac{1}{2}$

Prob $|1\rangle = 50\% = \frac{1}{2}$

Case-2 Gate matrix (H) State $|0\rangle$ Result V

Hadamard

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$|\psi|^2$ — Prob $|0\rangle = \frac{1}{2} = 50\%$.

Prob $|1\rangle = \frac{1}{2} = 50\%$

Case-3 Gate matrix (X) State $|1\rangle$ Result V

Pauli-X

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$|\psi|^2$ — Prob $|0\rangle = 1$

Prob $|1\rangle = 0$

Case-4 Gate Matrix (X) State $|0\rangle$ Result Vector

Pauli-X

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$|\psi|^2$ — Prob $|0\rangle = 0$

Prob $|1\rangle = 1$

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Case-5 Gate Matrix (Z) State $|1\rangle$ Result Vector

Pauli-Z

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$|\psi|^2$ — Prob $|0\rangle = 0 = 0\%$.

Prob $|1\rangle = 1 = 100\%$.

Case-6 Gate Matrix (Z) State $|0\rangle$ Result Vector

Pauli-Z

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

$|\psi|^2$ — Prob $|0\rangle = 1 = 100\%$.

Prob $|1\rangle = 0 = 0\%$

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* Borns postulate

- Express election with a wavefn $(\psi)(r)$
- What does wavefn represents?

→ Probability of finding an election at position r is proportional to the modulus square of the wave fn $|\psi|^2$

→ Born's postulate

- Using the squared modulus assures that we always have a real, positive quantity.

→ It also consistent with electromagnetic wave theory in which

→ Squared amplitude is the intensity of the electromagnetic wave &

→ Complex amplitude indicates the magnitude & phase of the filed.

- The wavefn itself is called probability amplitude.

→ No analogy is classical physics.

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॥ ॐ नमो भगवते वासुदेवाय नमः ॥

I) The wavefunction : Probability amplitude

In quantum mechanics, particles like electrons are not described by exact, deterministic trajectories, as they are in classical physics. Instead, they are described by a mathematical tool called a wave fn.

- Notation: Wavefn of an electron is $\Psi(\vec{r})$ where $\vec{r}$ represents the position vector in 3D space (x, y, z).
- Nature of the fn: The wavefunction itself is a complex valued fn meaning it involves imaginary nos i.
- Physical meaning: The Dide noter that $\Psi(x)$ is called probability amplitude. It is crucial to understand that the wavefn itself has no direct, measurable analogy in classical physics. You cannot measure a "probability amplitude" directly.

II) Born's Postulate — Physical Interpretation

Since a complex number cannot represent a physical probability — probabilities must be real numbers ~~with~~ bett 0 & 1, physicist Max Born proposed a rule to bridge the math & the physical world.

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- The postulate: Probability of finding an electron at a specific position r is proportional to the modulus square (or absolute square) of the wavefn at that position.

- Why the Modulus Square?

Squaring the absolute value of a complex number guarantees the result is a strictly real, positive quantity, which is a fundamental requirement for any probability.

→ Mathematics of Born's Rule

Let the wavefn be a complex number

$$\boxed{\psi = a + ib}$$

- To find the modulus square, we multiply the wavefunction by its complex conjugate denoted by ψ^* where sign of imaginary part is flipped

$$\boxed{\psi^* = a - ib}$$

$$\boxed{\rho = |\psi(r)|^2 = \psi^*(r)\psi(r)}$$

- This resulting value, $|\psi(r)|^2$ is known as probability density denoted by ρ.

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Calculating Actual probabilities

Because $|\psi(r)|^2$ is a density (probability per unit volume), the probability P of finding the electron within a specific, tiny volume element d^3r or dV

$$\boxed{dP = |\psi(r)|^2 \, d^3r}$$

To find the probability of the electron being in a larger specific region of space (volume V) you integrate the probability density over that volume:

$$\boxed{P = \iiint_V |\psi(r)|^2 \, dx\, dy\, dz}$$

III) ~~Not explicitly written in the bullets~~

The electron must exist somewhere in the universe, the total probability of finding it anywhere in all of space must be exactly 100% or 1.

The Normalization eqⁿ

$$\int_{-\infty}^{+\infty} |\psi(r)|^2 \, d^3r = 1$$

If a wave fn satisfies this equation, it is considered a valid, physical quantum state.

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IV) Electromagnetic wave analogy.

The slide draws a parallel to classical electromagnetism (EM) to make math more intuitive

- In EM Wave Theory: An electromagnetic wave has a ~~complete~~ complex electromagnetic field amplitude E.
This complex amplitude dictates the magnitude & phase of the wave, but it isn't what we see.
What we see or measure as light is the Intensity (I) which is square of amplitude.

$$I \propto |E|^2$$

- In Quantum mechanics: The wave fn is a complex amplitude containing magnitude & phase information about the quantum state.

The Intensity of this wave is probability density

$$\rho = |\Psi|^2$$

★ de Broglie's matter wave

De Broglie's matter wave hypothesis: $\lambda = \frac{h}{p}$

where $h = 6.626 \times 10^{-34}$ J.s,
Planck's constant.
experimentally confirmed by the electron diffraction experiments. — Davisson & JJ. Thompson
commonly used in crystallography.

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Wavelength $\lambda \propto \frac{1}{p \text{ (momentum)}}$

✡ Time Independent Schrodinger Eqⁿ

I) $\left[-\frac{\hbar^2}{2m}\nabla^2 + V(r)\right]\psi(r) = E\,\psi(r)$

- Problem is defined by the operator within the bracket.
  - Called Hamiltonian operator
  - Consistent of the term containing the differential operator ∇^2 and
  - Potential energy term, $V(r)$.
- Eigen value equation
  - Solves for both eigenvalue E and eigenfunction $\psi(r)$.
- TISE (Time Independent Schrodinger eqⁿ) is the fundamental equation in quantum physics used to find the allowed energy levels of a quantum system, like an electron in an atom, when potential Energy does not Δ with time.

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II) Hamiltonian operator ($\hat{H}$)

The slide notes that the specific physics problem is defined by the expression inside the square brackets. This entire term is called the Hamiltonian operator usually denoted by a "Hat"

$$\hat{H}\,\psi(r) = E\,\psi(r)$$

$\hat{H}$ = Total Energy of the system

$$\hat{H} = \hat{T} + \hat{V}$$

$\hat{T}$ = Kinetic Energy

$\hat{V}$ = Potential Energy

Here $\hat{H} = \underbrace{\frac{-\hbar^2}{2m}\nabla^2}_{\hat{T}} + \hat{V}(r)$

1) Kinetic Energy operator

$$\hat{T} = \frac{-\hbar^2}{2m}\nabla^2$$

$\hbar$ – Reduced planck's constant $\Rightarrow \hbar = \frac{h}{2\pi}$

It sets fundamental quantum scale.

m = mass of ~~electron~~ particle.

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- ∇^2 (Laplacian Operator): This is spatial differential operator. In a 3D cartesian coordinate system (x, y, z) it expands to second partial derivatives wrt space

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

2) Potential Energy Term ($V(r)$ or $\hat{V}$)

- $V(r)$: This function describes the environment the particle is in eg an electric field, a quantum well or attractive pull of a nucleus.

- r: The position vector in 3D space meaning the potential varies based on location (x, y, z) but not time (t).

III) Eigenvalue problem

TISE is Eigen value eqn

In linear algebra & differential eqn an eigenvalue fn takes the form

Operator [fn] = Constant [Same fn]

- Operator: Hamiltonian $(\hat{H})$

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- Eigenfn $\Psi(x)$: The quantum wave fn solving the differential eq^n gives you the valid wavefunctions for the system. the allowed states.

- Eigenvalue E: Total energy of the state

Sometimes we call E energy Eigenvalue.
Solving the equation yields specific discrete values for E. This is the mathematical origin of energy quantization why electrons can exist only in specific orbitals.

IV) 1D vs 3D forms

the 1D TISE

When movement is restricted to a single axis, the partial derivated of the laplacian simplify to a standard second derivate.

$$-\frac{\hbar^2}{2m}\frac{d^2\Psi(x)}{dx^2} + V(x)\,\Psi(x) = E\,\Psi(x)$$

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✱ 1D potential well problem

- Consider a 1D infinite potential well whose potential energy is given as

$$V(z) = \begin{cases} 0, & 0 \le z \le L \\ \infty, & \text{otherwise} \end{cases}$$

- outside the well; since potential is ∞ high,
  - Electrons cannot exist
  - We require the wavefunctions to be zero $\psi(z) = 0$

- Inside the well, Schrodinger eqⁿ is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dz^2} = E\psi(x)$$

- General solution is

$$\psi(z) = A\sin(kz) + B\cos(kz)$$

where $K = \sqrt{2mE/\hbar^2}$

- Requiring the wavefn to be continuous at the well boundaries

$$\psi(z=0) = \psi(z=L) = 0$$

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- We find $\psi(z) = A \sin(kz)$ with $k = n\pi/L$ where n is an integer
- Explicitly

$$\psi_n(z) = A_n \sin\left(\frac{n\pi z}{L}\right)$$

- Substituting back to Schrodinger's eq^n

$$\boxed{E_n = \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2}$$

n is Integer here.

- The constant A_n can be determined by normalization condition.
- From which, we find $A_n = \sqrt{\frac{2}{L}}$

— NOTE that A_n is arbitary within a factor $e^{i\theta}$

— Because $|\psi|^2$ is the physically meaningful quantity, not ψ itself.

☆ Full Solution: $\boxed{\psi_n(z) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi z}{L}\right)}$

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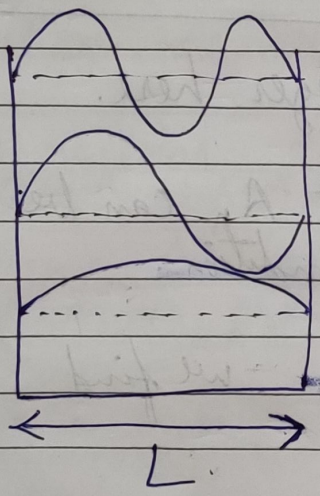
$$E_n = \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2$$

- Integer n is restricted to positive integers because
  - $n=0$ yields $\psi=0$ everywhere
  - A $-ve$ n yields the same ψ_n as its $+ve$ counterpart expect for the minus sign,
  - which is considered same wave fn



- Outside the well $z<0$ & $z>L$

$$\psi(z)=0$$

- Inside the well $0 \le z \le L$

$V=0$ (Potential Energy is zero)

TISE Eqn

$$\frac{-\hbar^2}{2m}\frac{d^2\psi}{dz^2} = E\psi(x)$$

Solving differential eqⁿ

$$\frac{d^2\psi}{dz^2} = \frac{-2mE}{\hbar^2}\psi$$

$$\frac{d^2\psi}{dz^2} + \frac{2mE}{\hbar^2}\psi = 0$$

defining wave number K.

$$K = \sqrt{\frac{2mE}{\hbar^2}} \Longrightarrow K^2 = \frac{2mE}{\hbar^2}$$

using value of K

$$\frac{d^2\psi}{dz^2} + K^2\psi = 0$$

General solution is

$$\boxed{\psi(z) = A\sin(Kz) + B\cos(Kz)}$$

A & B are constants of integration that must be determined by the physical boundaries of the system.

Applying boundary conditions

$$\psi(z=0) = \psi(z=L) = 0$$

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1st boundary condition $z = 0$

$$\psi(0) = A\sin(K(0)) + B\cos(K(0)) = 0$$

$$A(0) + B(1) = 0$$

$$\boxed{B = 0}$$

$$\psi(z) = A\sin(Kz)$$

2nd boundary condition $z = L$

$$\psi(L) = A\sin(KL) = 0$$

If $A = 0$ particle doesn't exist

trivial solution $\therefore$ sin term must be zero

$$\sin(KL) = 0$$

$$KL = n\pi$$

$$K = \frac{n\pi}{L}, \text{ where } n \text{ is an integer}$$

This leads to quantized wave fn

$$\boxed{\psi_n(z) = A_n \sin\left(\frac{n\pi z}{L}\right)}$$

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✡ Quantization of Energy

Because boundaries restricted K to specific discrete values
Energy E is also restricted to discrete values

$$K^2 = \frac{2mE}{\hbar^2}$$

using $K = \frac{n\pi}{L}$

$$\frac{n^2\pi^2}{L^2} = \frac{2mE}{\hbar^2}$$

$$\boxed{E_n = \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2}$$

✡ Normalization wave fn

$$\boxed{\int_0^L |A_n|^2 \sin^2\left(\frac{n\pi z}{L}\right) dz = 1}$$

using identity

$$\sin^2\theta = \frac{1-\cos(2\theta)}{2}$$

$$|A_n|^2 \int_0^L \frac{1-\cos\left(\frac{2n\pi z}{L}\right)}{2} dz = 1$$

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$$\frac{|A_n|^2}{2}\left[z - \frac{L}{2n\pi}\sin\left(\frac{2n\pi z}{L}\right)\right]_0^L = 1$$

Evaluating at limits L & 0

$$\frac{|A_n|^2}{2}\left[(L-0) - (0-0)\right] = 1$$

as $\sin(2n\pi) = 0$

$$\frac{|A_n|^2 L}{2} = 1$$

$$|A_n|^2 = \frac{2}{L}$$

$$\boxed{A_n = \sqrt{\frac{2}{L}}}$$

Final Solution

$$\psi_n(z) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi z}{L}\right)$$

Fully normalized Wave fn.